Julius AI Exploratory Analysis

Commerce Customer & Revenue Intelligence

Dataset: Online Retail Transaction Data (cleaned extract, MySQL-validated)
Analysis Period: December 1, 2010 – December 9, 2011
Records Analyzed: 392,692 line-level transactions
Tool Featured: Julius AI (AI-assisted exploratory data analysis)
Broader Toolchain: Python / Google Colab · Julius AI · SQL / MySQL · Power BI
Purpose: AI-assisted exploratory data analysis, pattern discovery, and business insight generation as one stage of an end-to-end retail analytics project.

Prepared as a Data Analyst portfolio artifact. All figures in this report are calculated directly from the underlying transaction dataset; results generated through Julius AI's exploratory workflow are presented alongside the validated calculations used to confirm them.

Table of Contents

1. Executive Summary

$8.89M Total Revenue	**18,532** Total Orders	**4,338** Unique Customers
$479.56 Avg. Order Value	**65.6%** Repeat Customer Rate	**82.0%** UK Revenue Share
65.2% Champions Revenue Share	**19.8%** Potentially Churned	**$554.8K** Churned-Customer Revenue

The analysis covers 392,692 line-level transactions spanning December 1, 2010 through December 9, 2011, representing 18,532 distinct orders placed by 4,338 unique customers across 37 countries and 3,665 distinct products, for a combined **$8,887,208.89** in total revenue and 5,152,002 units sold. The average order value across the period was **$479.56**. All figures in this summary are calculated directly from the underlying dataset.

Revenue is heavily concentrated in the United Kingdom. UK customers generated $7,285,024.64, or 81.97% of total revenue, from 3,920 of the 4,338 customers in the dataset. The remaining 36 countries combine for the balance, with the next-largest markets — the Netherlands, EIRE (Ireland), Germany, and France — each contributing under 3.5% individually.

A small group of best customers drives the majority of revenue. Using RFM (Recency, Frequency, Monetary) segmentation, the 957 customers classified as "Champions" (22.1% of the customer base) generated $5,791,640.74 — 65.17% of total revenue. This concentration is a core commercial finding: retaining and servicing this segment carries outsized weight relative to its size.

Repeat purchasing is the norm, but a third of customers never return. 65.58% of customers (2,845 of 4,338) placed more than one order, while 34.42% (1,493 customers) made only a single purchase in the observation window. Converting even a modest share of one-time buyers into repeat customers represents a clear, quantifiable growth lever.

Roughly one in five customers shows signs of inactivity. Using a 180-day-since-last-purchase threshold, 860 customers (19.82%) are classified as potentially churned, or inactive, at the close of the observation window. These customers collectively account for $554,832.97 in historical revenue (6.24% of total revenue) — value that is currently at risk of being lost if these relationships are not re-engaged.

Revenue accelerated sharply in the final quarter of the observation period. Monthly revenue rose from $644,051.04 in August 2011 to a peak of $1,156,205.61 in November 2011, an increase consistent with pre-holiday seasonal demand. December 2011 shows only $517,190.44 in revenue, but this reflects just 9 days of transaction data (December 1–9) and should not be read as a genuine decline.

Product-level revenue includes at least one significant anomaly. "PAPER CRAFT, LITTLE BIRDIE" ranks as the top product by revenue ($168,469.60) and by quantity (80,995 units), but the entire total is the result of a single order from a single customer on December 9, 2011 — not broad-based product demand. This transaction should be excluded or reviewed separately before drawing conclusions about organic product popularity.

2. Dataset Overview

The dataset used for this analysis is a cleaned extract of an online retail transaction log, provided as **online_retail_cleaned_mysql.csv**. It contains **392,692 transaction line records** across **11 columns**, covering the period **December 1, 2010 08:26 to December 9, 2011 12:50**. Each row represents a single product line within a customer invoice — one invoice can therefore span multiple rows. The dataset's core business entities are **customers** (identified by CustomerID), **products** (identified by StockCode/Description), **orders** (identified by InvoiceNo), and **countries** of sale. This file reflects the already-cleaned output of the project's Python / Google Colab preprocessing stage — see Section 3 for the data-quality checks performed to confirm this.

Column Reference

Column	Data Type	Business Meaning
InvoiceNo	Integer	Unique order/invoice identifier; groups line items into a single transaction.
StockCode	Text	Unique product code identifying the item sold.
Description	Text	Human-readable product name.
Quantity	Integer	Number of units of the product purchased on that line.
InvoiceDate	Datetime	Timestamp the order was placed.
UnitPrice	Decimal	Price per unit, in the dataset's base currency.
CustomerID	Integer	Unique identifier for the purchasing customer.
Country	Text	Country associated with the customer / shipment.
IsCancelled	Integer (0/1)	Flag indicating whether the order was cancelled (0 = not cancelled in this cleaned extract).
Revenue	Decimal	Line-level revenue, equal to Quantity × UnitPrice.
YearMonth	Text	Year-month bucket derived from InvoiceDate, used for time-series aggregation.

Scale Summary

Metric	Value
Transaction line records	392,692
Distinct orders (InvoiceNo)	18,532
Unique customers (CustomerID)	4,338
Unique products (StockCode)	3,665
Countries represented	37
Total units sold (Quantity)	5,152,002
Total revenue	$8,887,208.89
Date range	Dec 1, 2010 – Dec 9, 2011

3. Data Quality Assessment

A full data-quality pass was run against the provided extract. The table below summarizes the checks performed and their results.

Check	Result
Missing / null values (any column)	0 — no missing values in any of the 11 columns
Fully duplicated rows	0 duplicate records
Negative Quantity values	0
Zero Quantity values	0
Negative UnitPrice values	0
Zero UnitPrice values	0
Rows flagged IsCancelled = 1	0 (all records in this extract are completed, non-cancelled sales)
Invalid / unparseable InvoiceDate values	0
CustomerID completeness	100% populated (0 missing customer identifiers)

Interpretation

The absence of nulls, duplicates, cancellations, and negative or zero values indicates this extract has already passed through a cleaning stage (consistent with the Python / Google Colab preprocessing step in the broader project workflow — see Section 15). No further row-level cleaning was required before running the exploratory and validation analysis in this report.

Unusual Values Warranting Business Review

While the dataset is structurally clean, several individual values are statistical outliers that merit business review before being used to inform product or pricing decisions:

- **Extremely high quantity transaction:** a single order (Invoice 581483, December 9, 2011) for **80,995 units** of "PAPER CRAFT, LITTLE BIRDIE," placed by one customer. This single line accounts for the majority of that product's total lifetime quantity and revenue in the dataset.
- **Extremely high unit price:** a single line item under StockCode **POST** ("POSTAGE") carries a unit price of **$8,142.75** — far above the typical postage charge (median UnitPrice across the dataset is $1.95) and almost certainly a bulk shipping charge rather than a per-unit retail price.
- **PAPER CRAFT, LITTLE BIRDIE:** discussed above; despite ranking #1 by both revenue and quantity, performance is driven entirely by one transaction, not repeat product demand.
- **POST ("POSTAGE"):** a non-merchandise StockCode representing shipping/postage charges (1,099 orders, 331 customers, $77,803.96), not a physical product — should be excluded from product-performance rankings that are meant to reflect merchandise demand.
- **M ("Manual"):** a non-standard StockCode used for manually entered adjustments (279 line records, $53,419.93 in associated revenue) rather than catalog products.

These records likely represent bulk/wholesale orders, postage pass-through charges, or manual account adjustments rather than organic retail demand, and should be reviewed with the business before being included in product-performance or demand-forecasting conclusions.

Data Limitations

Because this extract contains only completed, non-cancelled positive-value sales (IsCancelled = 0 for all 392,692 rows), it does **not** support a full return, refund, or cancellation-rate analysis. Any cancelled or returned orders present in the original raw transaction log were removed prior to this extract being produced. Cancellation behavior should be analyzed separately against the raw, unfiltered transaction log if that business question needs to be answered.

4. Sales & Revenue Analysis

Figure 1. Monthly Revenue Trend

Monthly Revenue Detail

Month	Revenue	Orders
2010-12	$570,422.73	1,400
2011-01	$568,101.31	987
2011-02	$446,084.92	997
2011-03	$594,081.76	1,321
2011-04	$468,374.33	1,149
2011-05	$677,355.15	1,555
2011-06	$660,046.05	1,393
2011-07	$598,962.90	1,331
2011-08	$644,051.04	1,280
2011-09	$950,690.20	1,755
2011-10	$1,035,642.45	1,929
2011-11	$1,156,205.61	2,657
2011-12*	$517,190.44	778

**December 2011 contains only 9 days of data (Dec 1–9); it is not a complete calendar month.*

What the Chart Shows

Monthly revenue trends upward across the observation period, from $570,422.73 in the first partial month (December 2010) to a peak of $1,156,205.61 in November 2011. The strongest month on record is **November 2011**; the weakest **complete** calendar month is **February 2011**, at $446,084.92.

The most significant month-over-month movement is the run-up from August to November 2011: revenue grew 47.6% from August ($644,051.04) to September ($950,690.20), a further 8.9% into October ($1,035,642.45), and another 11.6% into November ($1,156,205.61) — a sustained three-month acceleration consistent with pre-holiday seasonal buying.

Important — December 2011 is incomplete. The dataset ends on December 9, 2011, so December's $517,190.44 reflects only 778 orders placed in the first 9 days of the month. The apparent month-over-month drop from November should not be interpreted as a genuine demand decline — it is a partial-month reporting artifact and should be excluded from trend or growth-rate conclusions.

5. Product Performance Analysis

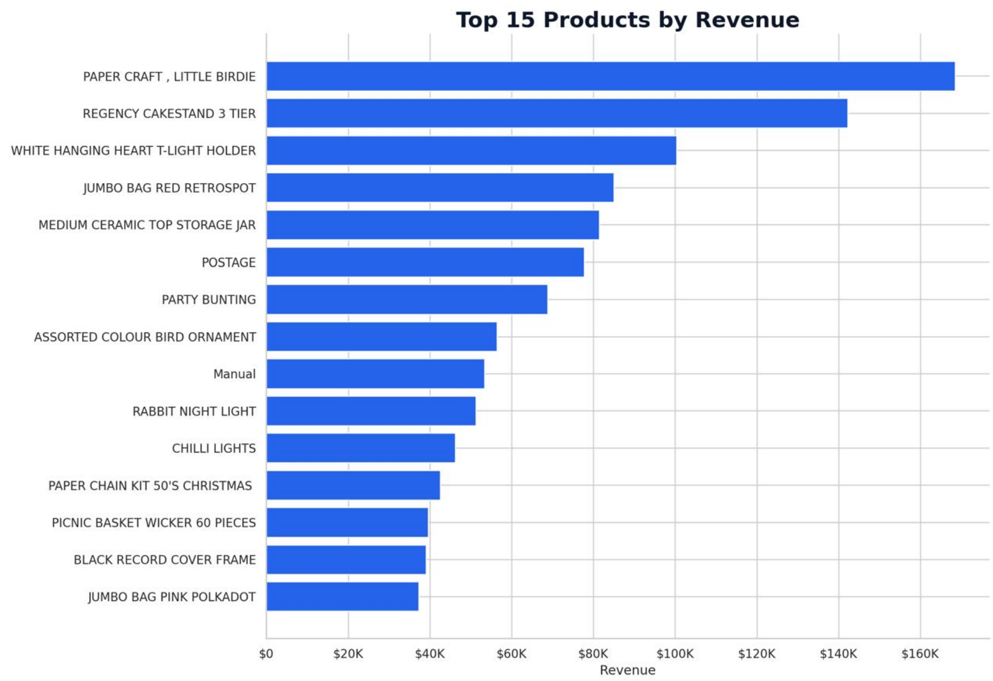

Figure 2. Top 15 Products by Revenue

Top Products by Revenue

Product	Revenue	Qty Sold	Orders	Customers
PAPER CRAFT, LITTLE BIRDIE	$168,469.60	80,995	1	1
REGENCY CAKESTAND 3 TIER	$142,264.75	12,374	1,703	881
WHITE HANGING HEART T-LIGHT HOLDER	$100,392.10	36,706	1,971	856
JUMBO BAG RED RETROSPOT	$85,040.54	46,078	1,600	635
MEDIUM CERAMIC TOP STORAGE JAR	$81,416.73	77,916	195	138
POSTAGE	$77,803.96	3,120	1,099	331
PARTY BUNTING	$68,785.23	15,279	1,379	708
ASSORTED COLOUR BIRD ORNAMENT	$56,413.03	35,263	1,375	678
Manual	$53,419.93	6,933	253	197
RABBIT NIGHT LIGHT	$51,251.24	27,153	801	450

The PAPER CRAFT, LITTLE BIRDIE Anomaly

"PAPER CRAFT, LITTLE BIRDIE" is the top-ranked product by both revenue ($168,469.60) and quantity (80,995 units) — but every one of those units was sold in a **single order to a single customer** on December 9, 2011. By contrast, the #3 product by revenue, WHITE HANGING HEART T-LIGHT HOLDER, generated $100,392.10 across 1,971 separate orders from 856 distinct customers — evidence of genuine, broad-based demand. This illustrates why **high quantity or high revenue alone does not indicate broad customer demand**: a product can top the leaderboard on the strength of one wholesale-style transaction.

A similar pattern appears with PICNIC BASKET WICKER 60 PIECES, which generated $39,619.50 in revenue from just 2 orders and 1 customer — another likely bulk or wholesale purchase rather than retail demand.

Recommended Approach to Evaluating Product Performance

Rather than ranking products on revenue or quantity alone, this analysis recommends evaluating products jointly across:

- Revenue
- Quantity sold
- Number of distinct orders
- Number of distinct customers
- Repeat purchase behavior across customers

Products that rank highly across order count and customer count (e.g., WHITE HANGING HEART T-LIGHT HOLDER, REGENCY CAKESTAND 3 TIER, JUMBO BAG RED RETROSPOT) represent genuine, diversified demand and are stronger candidates for inventory, marketing, and merchandising investment than single-transaction outliers.

6. Geographic Performance

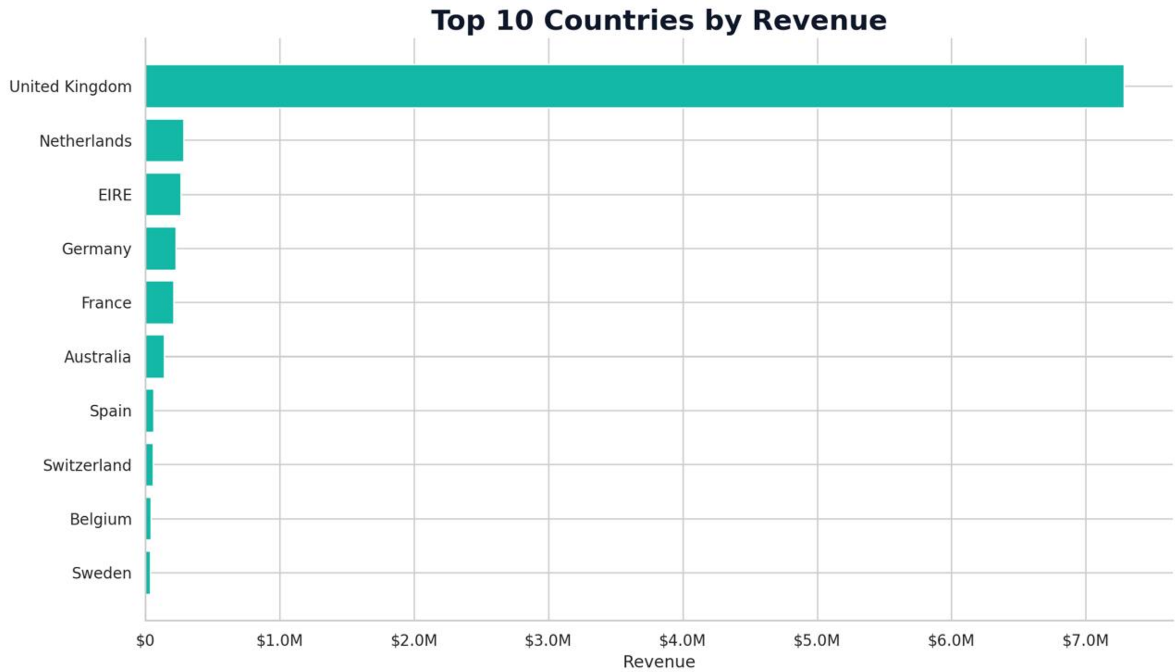

Figure 3. Top 10 Countries by Revenue

Country	Revenue	Customers	Revenue / Customer
United Kingdom	$7,285,024.64	3,920	$1,858.42
Netherlands	$285,446.34	9	$31,716.26
EIRE (Ireland)	$265,262.46	3	$88,420.82
Germany	$228,678.40	94	$2,432.75
France	$208,934.31	87	$2,401.54
Australia	$138,453.81	9	$15,383.76
Spain	$61,558.56	30	$2,051.95
Switzerland	$56,443.95	21	$2,687.81
Belgium	$41,196.34	25	$1,647.85
Sweden	$38,367.83	8	$4,795.98

United Kingdom Dominance

The United Kingdom accounts for **$7,285,024.64** in revenue — **81.97%** of the dataset's total — from 3,920 of the 4,338 total customers (90.4% of the customer base). This makes the UK both the largest market by revenue and the most diversified, with a moderate revenue-per-customer figure of $1,858.42.

Geographic Concentration Risk: Netherlands, EIRE, and Australia

Three markets show a materially different pattern from the UK: high revenue relative to a very small number of customers. **EIRE (Ireland)** generated $265,262.46 from just **3 customers** ($88,420.82 revenue per customer). The **Netherlands** generated $285,446.34 from **9 customers** ($31,716.26 per customer). **Australia** generated $138,453.81, also from **9 customers** ($15,383.76 per customer). In each case, a handful of accounts — likely wholesale, distributor, or B2B-style buyers rather than typical retail customers — are responsible for the entire market's revenue.

This concentration is a risk, not automatically an opportunity. **This analysis does not conclude that these markets should be expanded.** Before treating the Netherlands, EIRE, or Australia as growth targets, the business should investigate: the account type behind these customers (wholesale vs. retail), true margins after shipping and currency costs, the concentration risk of relying on 3–9 accounts for six-figure regional revenue, and whether repeat behavior from these accounts is stable or one-off.

7. Customer Behavior Analysis

Figure 4. Executive Overview — Revenue, Customer Distribution & Activity Status

Metric	Value
Total unique customers	4,338
Repeat customers (2+ orders)	2,845 (65.58%)
One-time customers (1 order)	1,493 (34.42%)
Average revenue per customer	$2,048.69
Median revenue per customer	$668.57
Highest single-customer revenue	$280,206.02 (Customer 14646)

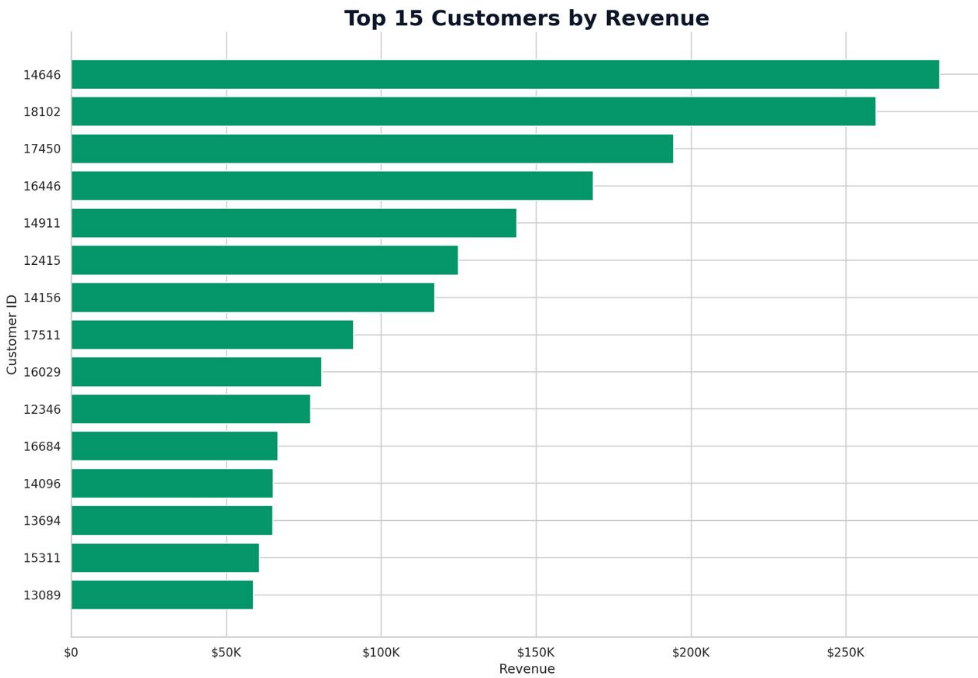

Figure 5. Top 15 Customers by Revenue

Customer ID	Total Revenue
14646	$280,206.02
18102	$259,657.30
17450	$194,390.79
16446	$168,472.50
14911	$143,711.17

Loyal vs. Bulk vs. One-Time Customers

The customer base splits into three commercially distinct groups. **Loyal repeat customers** place multiple, typically smaller orders over time (2,845 customers fall into this pattern) — this is the healthiest and most predictable revenue base. **High-value bulk customers**, such as Customer 12346 ($77,183.60 from a single order) or the wholesale-style country accounts discussed in Section 6, generate large revenue from very few transactions; their value is real but far less predictable and carries concentration risk. **One-time customers** (1,493 of 4,338, or 34.42%) have not returned within the observation window and represent unrealized lifetime value.

Why Customer Concentration Matters

The top 15 customers alone contribute a substantial share of total revenue (over $1.7M combined across the customers shown in Figure 5), out of 4,338 total customers. Revenue this concentrated in a small number of accounts increases retention risk: losing even a handful of top accounts would have an outsized effect on total revenue, which is why Section 11 (Churn & Retention) and Section 14 (Recommendations) both prioritize individual monitoring of high-value accounts rather than only broad-based retention campaigns.

8. Customer Distribution by Number of Orders

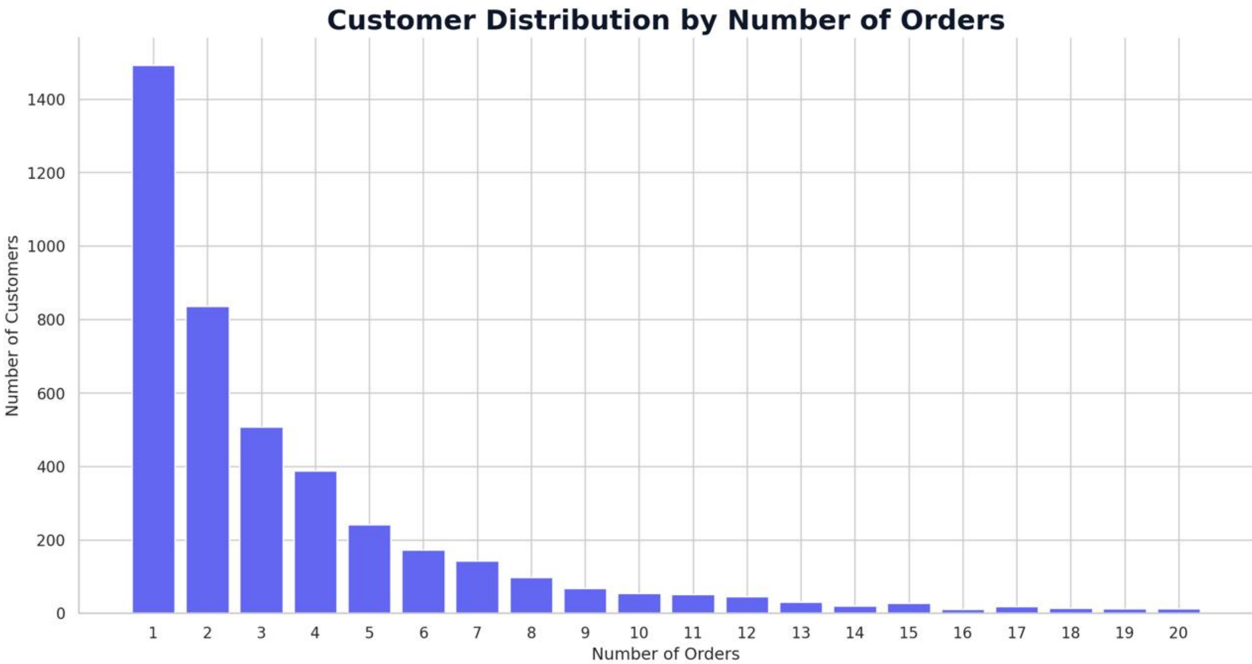

Figure 6. Customer Distribution by Number of Orders

How Customers Are Distributed by Order Frequency

Number of Orders	Customers
1	1,493
2	835
3	508
4	388
5	242
6	172
7	143
8	98
9	68
10	54
11–20 (combined)	212
21+ (combined)	95

The distribution is heavily right-skewed: **1,493 customers (34.42%)** — the largest single group — placed exactly one order, and the number of customers falls off sharply with each additional order after that (835 customers with 2 orders, 508 with 3, and so on). A small tail of 95 customers placed more than 20 orders each, with the single most frequent customer placing 209 distinct orders.

Importance of Repeat Purchasing

Because 65.58% of customers do return for a second order or more, repeat purchasing is the norm rather than the exception in this business — but the one-time customer group is still large enough (1,493 customers) to represent a meaningful growth opportunity. Converting even a modest fraction of one-time buyers into second-time buyers would shift a large number of customers into the far more valuable repeat-purchase segment, where average customer revenue is materially higher than the single-order segment.

9. RFM Analysis

RFM (Recency, Frequency, Monetary) segmentation was used to classify customers by purchase behavior. The three dimensions are:

Dimension	Definition
Recency	Number of days between a customer's most recent purchase and the analysis snapshot date (December 10, 2011, the day after the last transaction in the dataset).
Frequency	Number of distinct orders (unique InvoiceNo values) placed by the customer.
Monetary	Total historical revenue generated by the customer across all orders.

Each customer was scored 1–5 on each dimension using quintile-based scoring (5 = best), then assigned to one of five behavioral segments based on the combination of scores:

Segment	Definition
Champions	High recency, high frequency, and high monetary value — the most engaged, most valuable, most recently active customers.
Loyal	Good recency and consistent order frequency; regular, dependable repeat buyers.
Potential Loyalists	Recently active but with limited order history so far — early-stage customers with upside if nurtured.
At Risk	Previously frequent or valuable customers whose recency has declined — a warning sign, not yet full churn.
Hibernating	Low recency, frequency, and monetary value — largely inactive customers with limited recent engagement.

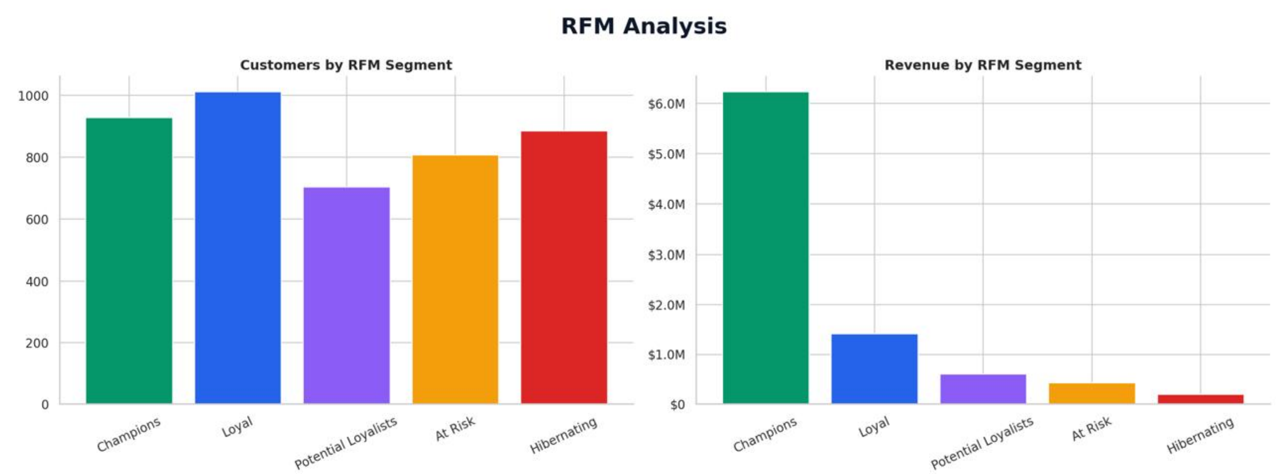

Figure 7. RFM Analysis — Customers and Revenue by Segment

Segment Sizes and Revenue

Segment	Customers	% of Customers	Revenue	% of Revenue
Champions	957	22.06%	$5,791,640.74	65.17%
Loyal	1,003	23.12%	$1,472,865.64	16.57%
At Risk	643	14.82%	$798,052.51	8.98%
Hibernating	1,416	32.64%	$679,430.26	7.65%
Potential Loyalists	319	7.35%	$145,219.74	1.63%

Segment sizes and revenue totals are the validated, calculated result (Section 16 explains how this calculation relates to the exploratory chart above).

10. Revenue by RFM Segment

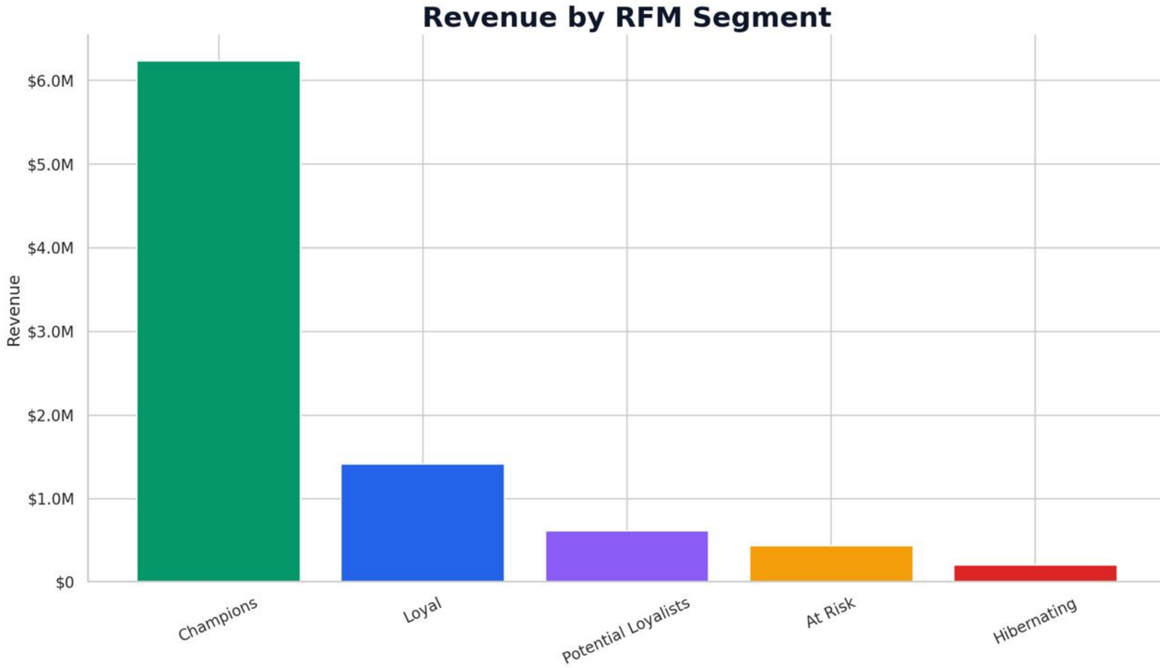

Figure 8. Revenue by RFM Segment

Champions Drive the Majority of Revenue

The **Champions** segment contributes the most revenue by a wide margin: **957 customers** (22.06% of the customer base) generated **$5,791,640.74**, or **65.17% of total revenue**. The **Loyal** segment is the second-largest revenue contributor at $1,472,865.64 (16.57%), from a slightly larger group of 1,003 customers. Combined, Champions and Loyal customers — just 45.2% of the customer base — account for **81.7%** of total revenue.

At Risk Segment

The **At Risk** segment contains **643 customers** who have historically generated **$798,052.51** (8.98% of total revenue) but whose recency has declined. These are customers who were previously frequent or valuable purchasers and are now showing early warning signs of disengagement — distinct from the Hibernating segment, which reflects customers who were never particularly engaged to begin with.

Why High-Value Inactive Customers Deserve Special Attention

Because revenue is so concentrated in the Champions segment, the financial impact of losing even a small number of high-value customers is disproportionately large compared to losing an equivalent number of Hibernating customers. A targeted, individual-account approach to monitoring At Risk and high-value inactive customers (see Section 11) is a more efficient use of retention resources than broad, undifferentiated re-engagement campaigns.

11. Churn & Retention Analysis

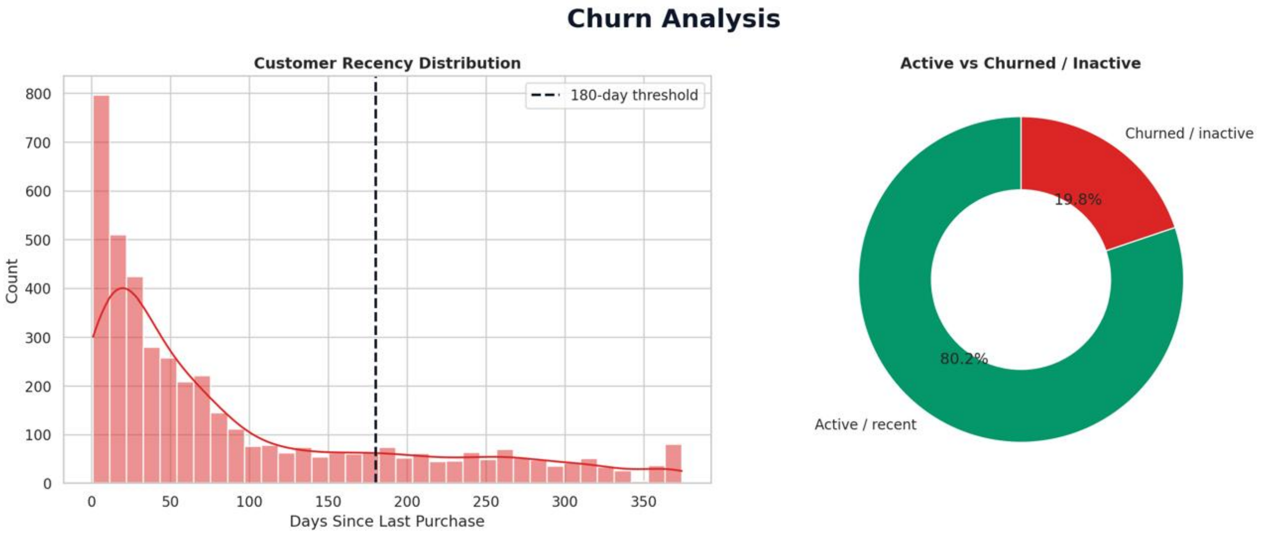

Figure 9. Churn Analysis — Recency Distribution & Active vs. Inactive Split

Methodology

Consistent with the exploratory analysis, this report classifies a customer as **potentially churned** if more than **180 days** have passed since their most recent purchase, measured against the dataset's snapshot date of December 10, 2011 (one day after the last transaction in the dataset).

Important — this is not confirmed churn. This report deliberately uses the terms "Potential Churn" or "Customer Inactivity Indicator" rather than "confirmed churn." The dataset ends on December 9, 2011 and provides no visibility into whether these customers made purchases after that date. A 180-day recency threshold is a reasonable, commonly used proxy for disengagement, but it cannot distinguish a customer who has permanently left the business from one who simply has an infrequent, longer purchase cycle.

Metric	Value
Active customers (≤180 days since last purchase)	3,478 (80.18%)
Potentially churned / inactive customers (>180 days)	860 (19.82%)
Historical revenue from potentially churned customers	$554,832.97
Potentially churned revenue as % of total revenue	6.24%

High-Value Inactive Customers

Several potentially churned customers carry disproportionately high historical value and warrant individual review rather than generic win-back treatment:

Customer ID	Days Since Last Purchase	Orders Placed	Historical Revenue
12346	326	1	$77,183.60
15749	235	3	$44,534.30

Customer ID	Days Since Last Purchase	Orders Placed	Historical Revenue
15098	182	3	$39,916.50
12590	211	2	$9,864.26
13093	276	8	$7,832.47

Customer 12346 is a notable case: a single $77,183.60 order (making them the 10th-highest revenue customer overall) with no repeat purchase in 326 days — the profile of a high-value, one-time or wholesale buyer rather than a lapsed regular customer, and worth understanding before designing a retention approach.

12. Pattern & Anomaly Discovery

Late-Year Revenue Acceleration

Observation: Revenue rose 47.6% from August to September 2011, then continued climbing through November.

Why it matters: Signals a strong seasonal / pre-holiday demand pattern that should inform inventory and staffing planning.

What should be investigated: Whether this pattern repeats in prior years' data (if available) to confirm it is seasonal rather than one-off growth.

November Peak

Observation: November 2011 is the single highest-revenue month in the dataset at $1,156,205.61.

Why it matters: Establishes the commercial ceiling and seasonal peak for planning purposes.

What should be investigated: Which specific product categories and customer segments drove the November peak, to replicate it.

Incomplete December Data

Observation: December 2011 shows only $517,190.44 in revenue, but covers just 9 of 31 days.

Why it matters: A naive month-over-month comparison would incorrectly read this as a steep decline.

What should be investigated: Extending the dataset through a full December to get a true reading of year-end performance.

High-Volume Product Anomaly

Observation: PAPER CRAFT, LITTLE BIRDIE tops both revenue and quantity rankings from a single 80,995-unit order.

Why it matters: Product leaderboards built on raw revenue or quantity can be misleading without order/customer counts.

What should be investigated: Whether this was a legitimate wholesale sale, a data-entry error, or a return/adjustment miscoded as a sale.

High-Value Customer Concentration

Observation: The top 15 customers by revenue range from $58,762 to $280,206 each, while median customer revenue is $668.57.

Why it matters: A small number of accounts represent outsized revenue risk and reward.

What should be investigated: Account-level retention plans and contract terms for the top 15–20 customers specifically.

Geographic Concentration

Observation: EIRE ($88,420.82/customer), the Netherlands ($31,716.26/customer), and Australia ($15,383.76/customer) each derive high revenue from single-digit customer counts.

Why it matters: Regional revenue in these markets depends on a handful of accounts, not a diversified customer base.

What should be investigated: The account type (wholesale/distributor vs. retail) and contract stability behind these customers.

High-Revenue Customers with Low Order Frequency

Observation: Customer 12346 generated $77,183.60 from a single order; several other high-value customers show few orders relative to their revenue.

Why it matters: High Monetary scores do not always correspond to high Frequency — these customers behave differently from typical loyal repeat buyers.

What should be investigated: Whether these are wholesale/B2B accounts that should be tracked and managed separately from retail customers.

Potential Wholesale / Bulk Purchasing Behavior

Observation: Products like PAPER CRAFT, LITTLE BIRDIE and PICNIC BASKET WICKER 60 PIECES show high revenue from extremely few orders and customers.

Why it matters: Mixing wholesale-style transactions into retail product-performance analysis can distort demand signals.

What should be investigated: Whether a separate wholesale/B2B reporting category should be created to isolate these transactions.

13. Key Business Insights

Insight 1 — Revenue Concentration in Champions

Observation: Champions (957 customers, 22.06% of the base) generate 65.17% of total revenue ($5,791,640.74).

Business Meaning: A small, identifiable group of customers is responsible for the large majority of the business's revenue — protecting this segment should be a top operational priority.

Insight 2 — UK Market Dominance

Observation: The United Kingdom generates $7,285,024.64, or 81.97% of total revenue, from 90.4% of all customers.

Business Meaning: The business is fundamentally a UK-first retailer; international revenue, while present, is a small share of the total.

Insight 3 — Meaningful Repeat Purchase Base

Observation: 65.58% of customers (2,845 of 4,338) placed more than one order.

Business Meaning: Repeat purchasing is the norm, indicating reasonable product-market fit and customer satisfaction with the core offering.

Insight 4 — Large One-Time Customer Pool

Observation: 34.42% of customers (1,493) made only a single purchase during the observation window.

Business Meaning: This is the single largest addressable growth opportunity in the dataset: converting even a modest share of these customers into repeat buyers would materially grow the loyal customer base.

Insight 5 — At Risk Segment Represents Recoverable Revenue

Observation: 643 customers classified At Risk have generated $798,052.51 historically but show declining recency.

Business Meaning: These customers were previously engaged and valuable — they are more efficient to win back than acquiring new customers from scratch.

Insight 6 — Potential Churn Affects One in Five Customers

Observation: 19.82% of customers (860) have not purchased in over 180 days, representing $554,832.97 in historical revenue.

Business Meaning: This inactive base, while not confirmed lost, represents revenue at risk that should be prioritized for re-engagement before it is permanently lost.

Insight 7 — Product Leaderboard Distortion

Observation: PAPER CRAFT, LITTLE BIRDIE ranks #1 by revenue and quantity from a single order/customer, unlike genuinely popular products such as WHITE HANGING HEART T-LIGHT HOLDER (1,971 orders, 856 customers).

Business Meaning: Raw revenue or quantity rankings can mask whether performance reflects broad demand or a single transaction — order and customer counts must be reviewed alongside revenue.

Insight 8 — Geographic Revenue Concentration Outside the UK

Observation: EIRE, the Netherlands, and Australia each generate five-to-six-figure revenue from fewer than 10 customers per country.

Business Meaning: International revenue outside the UK is currently dependent on a small number of large accounts rather than a broad customer base, creating concentration risk.

Insight 9 — Strong Seasonal Pattern in Late 2011

Observation: Revenue grew from $644,051.04 in August to $1,156,205.61 in November 2011 — a 79.5% increase over three months.

Business Meaning: The business exhibits a clear, sizeable holiday-season demand curve that should shape inventory, staffing, and marketing timing.

Insight 10 — Top Customer Concentration

Observation: The single highest-revenue customer (ID 14646) generated $280,206.02 — more than 420 times the median customer's $668.57.

Business Meaning: Revenue is extremely right-skewed at the customer level; average revenue per customer ($2,048.69) is not representative of a typical customer.

Insight 11 — Non-Merchandise Line Items Affect Product Analysis

Observation: POSTAGE (StockCode POST) and Manual (StockCode M) entries total $131,223.89 combined in revenue but are not physical products.

Business Meaning: These should be excluded from merchandise-focused product performance analysis to avoid overstating the popularity of any specific product category.

Insight 12 — December 2011 Cutoff Limits Full-Period Conclusions

Observation: The dataset ends 9 days into December 2011, leaving the most recent recency window artificially inflated for many customers.

Business Meaning: Churn and recency figures should be re-run once a complete, more recent dataset is available, as the current 19.82% churn figure may shift once the observation window is extended.

14. Business Recommendations

Customer Retention

- Target the 643 At Risk customers with re-engagement offers before they cross the 180-day inactivity threshold and move into the Hibernating segment.
- Build a second-purchase campaign specifically aimed at the 1,493 one-time customers, since converting even 10–15% of this group into repeat buyers would meaningfully grow the Loyal segment.
- Re-run the 180-day churn analysis on a rolling basis once more recent transaction data is available, since the current dataset's December 9, 2011 cutoff likely inflates the near-term inactive count.

High-Value Customer Management

- Assign individual account monitoring to the top 15–20 customers by revenue (e.g., Customers 14646, 18102, 17450), given their outsized contribution relative to the median customer.
- Review high-value inactive customers such as Customer 12346 ($77,183.60 from one order, 326 days inactive) individually rather than through mass-market win-back campaigns.
- Protect the Champions segment specifically — losing even a small percentage of these 957 customers would have a disproportionate revenue impact given their 65.17% revenue share.

Product Strategy

- Investigate whether the 80,995-unit PAPER CRAFT, LITTLE BIRDIE order represents legitimate wholesale demand, a data-entry error, or a miscoded transaction before using this product to inform inventory or marketing decisions.
- Exclude POSTAGE and Manual (non-merchandise) line items from product-performance rankings used for merchandising decisions.
- Prioritize inventory and marketing investment in products with both high revenue and high order/customer counts (e.g., WHITE HANGING HEART T-LIGHT HOLDER, REGENCY CAKESTAND 3 TIER) over single-transaction outliers.

Marketing Strategy

- Design a distinct marketing motion for Potential Loyalists (319 customers) to accelerate their transition into the Loyal or Champions segments while their engagement is still recent.
- Time major promotional and inventory pushes around the observed August–November seasonal ramp, given the 79.5% revenue growth seen over that window in 2011.
- Avoid broad, undifferentiated campaigns for the Hibernating segment (1,416 customers, 7.65% of revenue); prioritize spend on At Risk and Potential Loyalist segments where the return on re-engagement is likely higher.

Geographic Strategy

- Evaluate the Netherlands, EIRE, and Australia using revenue-per-customer, account type, and repeat rate before treating them as expansion targets, given their reliance on fewer than 10 customers each.

- Continue prioritizing the UK market operationally, since it represents 81.97% of revenue and 90.4% of the customer base — the largest lever for near-term revenue growth.
- Investigate margins and shipping costs for the concentrated international accounts before scaling similar wholesale-style relationships in new markets.

Revenue Growth

- Focus retention resources on the combined Champions + Loyal segments (45.2% of customers, 81.7% of revenue) as the highest-leverage group for sustaining revenue.
- Quantify the revenue opportunity of converting one-time customers to repeat status, using the $2,048.69 average customer revenue as a benchmark for potential uplift per converted customer.
- Extend the dataset beyond December 9, 2011 to validate whether the November peak and the churn/inactivity figures hold once a complete, more recent period is analyzed.

15. How Julius AI Was Used

Julius AI was used as an **AI-assisted exploratory analysis tool** during the early stage of this project, to accelerate pattern discovery and generate a first-pass set of visualizations and hypotheses ahead of deeper, structured analysis. Specifically, Julius AI supported:

- Dataset exploration and initial profiling of the retail transaction extract
- Data-quality assessment (missing values, duplicates, outliers, cancellation flags)
- Exploratory data analysis across revenue, product, and customer dimensions
- Revenue trend analysis (monthly revenue, seasonality)
- Product performance analysis (top products by revenue and quantity)
- Customer behavior analysis (repeat vs. one-time customers, revenue distribution)
- RFM exploration (recency, frequency, monetary scoring and segmentation)
- Churn / inactivity analysis (recency-threshold-based customer status)
- Pattern and anomaly discovery (the PAPER CRAFT outlier, geographic concentration, etc.)
- Business insight generation from the exploratory outputs
- Visualization recommendations for communicating findings

Julius AI was one stage of a broader analytics workflow — not a replacement for it. Julius AI did not independently build the complete project. The end-to-end workflow combined multiple tools, each used for what it does best:

Stage	Tool	Role
1	Python / Google Colab	Data cleaning and preparation of the raw transaction log into the analysis-ready extract used in this report.
2	Julius AI	AI-assisted exploration and pattern discovery — generating the first-pass charts and hypotheses summarized in this report.
3	SQL / MySQL	Structured business analysis and validation of key metrics against the cleaned dataset.
4	Power BI	DAX-based measures, interactive dashboards, and business storytelling for stakeholder-facing delivery.

16. Validation & Analytical Workflow

Findings generated during the Julius AI exploratory pass were not treated as unquestionable outputs. For this report, every headline metric was independently recalculated directly from the underlying **online_retail_cleaned_mysql.csv** dataset — the same validation discipline the project applies through its SQL and Power BI stages — rather than being taken as-is from the exploratory charts.

Where Validation Confirmed the Exploratory Findings

The recalculated figures closely match the exploratory results across every major metric reviewed, including the churn rate (19.82% recalculated vs. 19.8% shown in the exploratory chart), the active/inactive customer split, the monthly revenue trend and seasonal shape, the top products by revenue, the top countries by revenue, the top customers by revenue, and the customer order-frequency distribution — all of which matched to the underlying data exactly.

Where Validation Refined the Exploratory Findings

One area showed a modest difference: the RFM segment revenue split. The exploratory Julius AI chart (Figure 7) grouped customers using its own internal quintile-scoring pass, while the validated recalculation in this report (Section 9–10) applied an explicitly documented rule set (recency/frequency/monetary quintile scoring against defined segment thresholds — see Section 9). Both approaches agree directionally — Champions is the clear majority revenue driver in both — but the exact customer counts and revenue splits per segment differ slightly (for example, the validated Champions revenue share is 65.17%, close to but not identical to the exploratory chart's visual proportions). This is an expected and normal outcome of RFM analysis, where segment boundaries are threshold-sensitive; the validated figures in Sections 9–10 are the authoritative numbers for this report.

Validation Steps Performed

- Row counts, column structure, and date range were confirmed directly against the source CSV.
- Revenue, order, customer, and product totals were recalculated with independent aggregation logic and cross-checked against the exploratory chart values.
- Customer-level calculations (RFM scores, churn status, order counts) were computed directly from transaction-level data rather than taken from chart estimates.
- Country-, product-, and customer-level rankings were recomputed and matched against the corresponding exploratory visualizations before being included in this report.

Relevant findings were validated against the project's SQL / Python / Power BI calculations where applicable, consistent with the broader analytical workflow described in Section 15.

17. Visualization Index

Analysis	Visualization Type	File
Monthly Revenue Trend	Line Chart	monthly_revenue_trend.png
Executive Overview (Revenue, Distribution, Activity)	4-Panel Dashboard	executive_overview.png
Top 15 Products by Revenue	Horizontal Bar Chart	top_15_products_by_revenue.png
Churn Analysis (Recency Distribution + Active/Inactive)	Histogram + Donut Chart	churn_analysis.png
RFM Analysis Overview (Customers & Revenue by Segment)	2-Panel Bar Chart	rfm_analysis_overview.png
Revenue by RFM Segment	Bar Chart	revenue_by_rfm_segment.png
Top 10 Countries by Revenue	Horizontal Bar Chart	top_10_countries_by_revenue.png
Customer Distribution by Number of Orders	Column Chart	customer_distribution_by_orders.png
Top 15 Customers by Revenue	Horizontal Bar Chart	top_15_customers_by_revenue.png

18. Final Conclusion

This analysis of 392,692 transactions across 4,338 customers and 37 countries shows a fundamentally healthy, UK-centered retail business generating $8,887,208.89 in revenue over a 12-month window, with a clear seasonal peak in late 2011 and a majority-repeat customer base (65.58% repeat purchase rate).

The single most important structural finding is revenue concentration: the Champions segment — just 957 customers — generates 65.17% of all revenue, and the top 15 customers alone account for a disproportionate share of total value. Protecting and individually managing this small group of high-value relationships should be the top commercial priority to emerge from this analysis.

At the same time, two clear growth levers stand out: the 1,493 one-time customers represent an untapped conversion opportunity, and the 643 At Risk customers — previously valuable, now showing declining engagement — represent recoverable revenue that is more efficient to win back than acquiring new customers. The 19.82% potential churn rate, while not confirmed churn, is a meaningful signal that should be monitored on a rolling basis as more recent data becomes available.

Several product and geographic anomalies — most notably the single-transaction PAPER CRAFT, LITTLE BIRDIE order and the customer-concentrated Netherlands, EIRE, and Australia markets — are reminders that raw revenue and quantity rankings can be misleading without also examining order counts and customer counts.

Finally, this project illustrates the practical value of combining AI-assisted exploratory tools with traditional analytics infrastructure: Julius AI accelerated the discovery of these patterns, but every figure presented in this report was independently validated directly against the source dataset, and the findings are intended to be carried forward into SQL-based business analysis and Power BI dashboards for ongoing monitoring — the combination of fast AI-assisted discovery and rigorous validation is what makes the findings in this report reliable enough to act on.